

MANAGEMENT & TECHNICAL REPORT

PROJECT TITLE: Predictive Customer Retention and Churn
Predictor
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EXECUTIVE SUMMARY

Legacy customer retention models in the aviation sector have traditionally relied on static, historical Customer Lifetime Value (CLV) accounting and reactive demographic tracking. This leads to significant financial exposure by failing to identify behavioral friction before customer churn occurs. This report explains the development and deployment of a predictive ML framework made to solve this vulnerability from a database of over 16,000 members.

A Random Forest Classifier was deployed on the cleaned database. My analysis reveals a critical disconnect:

- Wealth, the most obvious factor of customer retention correlates **poorly** with actual brand loyalty
- Aggressive promotional sign-ups hide the highly unstable churn distributions, even among elite tier cardholders.

To make use of these insights, a decision-support interface was constructed. This platform bridges the gap between predicting future data and marketing execution. It enables administrators with no technical knowledge to run real-time "What-If" simulations, evaluate their marketing campaign budgets against various customer retention parameters, and be able to export it offline instantly. This interface helps the operations team gather an idea on deploying specific, metric-driven retention strategies that protect the revenue and stabilize long-term cohort momentum.

1. THE VALUE DISCONNECT

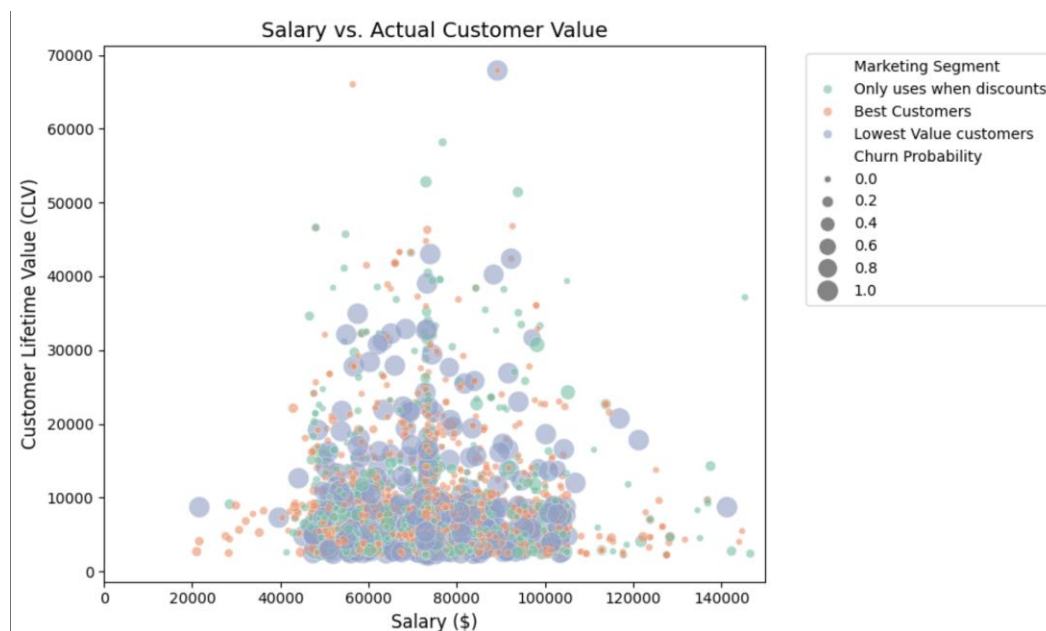
1.1 PROBLEM BOUNDARY

Modern airline loyalty ecosystems have become a core revenue source for aviation companies. But accessing the account's health using static metrics creates a false sense of customer stability.

The true customer value is dynamic. It is driven by short term modifications and ecosystem interactions rather than the accumulation of lifetime's worth of customer statistics. When a high-tier customer activity slows down, the old legacy systems in place are unable to flag the account early and only do so when the relationship is unrecoverable. This delay causes the loss of millions of dollars in future revenue to churn.

1.2 THE DEMOGRAPHIC-WEALTH MIRAGE

The EDA of our customer database exposed a massive disconnect between consumer demographics and actual value. Plotting customer income brackets directly against the historical CLV highlights that top-tier income segments frequently contain dense clusters of low value unengaged users. So, high personal wealth does not, in any way, translate to likeness for the brand.



The visual distribution shows that high-earning individuals populate every layer of the risk spectrum. Behavioral engagement metrics completely override these raw wealth indicators, because of which targeted retention algorithms must prioritize transaction momentum indicators over such static demographics.

2. ML FRAMEWORK

2.1 MODEL ARCHITECTURE & ENGINEERING PIPELINE

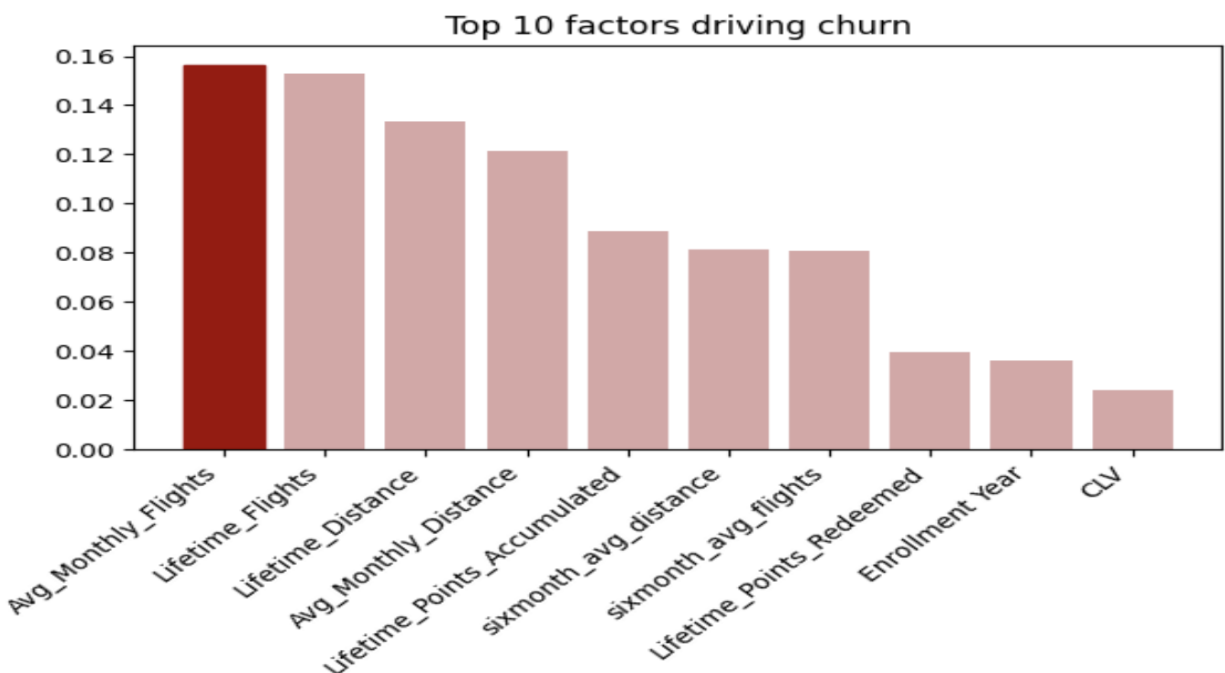
To calculate the churn risk for each individual customer, I made a robust model which utilizes a Random Forest Classifier. This model was selected for its ability to capture non-linear relationships across high-cardinality nominal values and binary indicators without the need of heavy feature scaling.

The model has been trained on a database of over 16,000 customers. Feature engineering focused heavily on generating high signal behavioral tracking columns, like:

1. **FLIGHT MOMENTUM:** It is the ratio of the last 6-month travel frequency compared to that customer's lifetime historical monthly average
2. **POINT BURN INERTIA:** Split the account balance by lifetime redemption frequency to show inactive accounts who are only hoarding points
3. **ACQUISITION VECTORS:** Encoding binary variables to isolate organic sign-ups from those which are part of promotional campaigns (Enrollment Type_Standard column)

2.2 MODEL DIAGNOSTIC EVALUATION

The model's performance was evaluated using rigorous stratification splits to prevent any training leak and account for minor class imbalances.

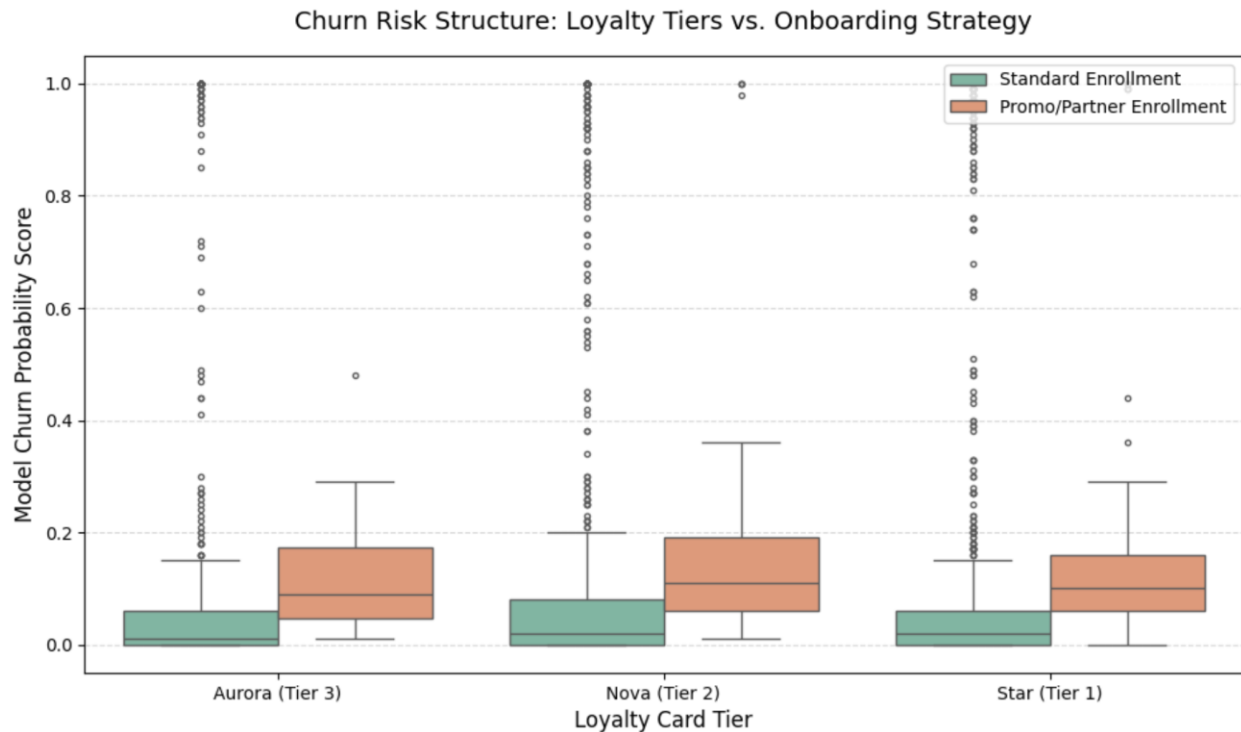


As illustrated in this bar chart, short term behavioral indicators hold more predictive power than static demographics. By tuning our classification thresholds to favor a **RECALL PROFILE**, the model ensures that the business flags critical revenue anomalies **EARLY**, minimizing the financial impact of unpredicted churn events.

3. CORE CUSTOMER SEGMENTATION

3.1 ONBOARDING CHANNEL AS A CORE RISK VECTOR

A key finding of this report has been the direct relationship between a customer's source of loyalty tier acquisition and their long-term loyalty. Cross referencing these loyalty tiers with their onboarding methods exposed major risk variations across identical card brackets.



This distribution highlights a very simple pattern:

- **STANDARD ENROLLMENT:** Shows tightly squeezed risk distributions right at the bottom baseline (risk scores of 0-0.2) for all tiers. These members have organically generated loyalty towards the brand and hence provide a highly stable revenue stream.
- **PROMO/PARTNER ENROLLMENT:** Shows volatile risk spreads stretching all the way up to a 100% churn risk. These accounts are highly prone to dropping off once the source of incentive is over.

3.2 HIGH-TIER RETENTION PARADOX

This risk distribution exposes a major vulnerability amongst our most prestigious members. Customers holding elite status (Aurora (Tier 3)) who were acquired via promotional campaigns exhibit an unexpected risk distribution. Despite having access to the best possible rewards, their risk metrics behave similarly to lower-tier promotional accounts.

Hence, higher tier alone doesn't guarantee brand retention if the customer lacks organic interest during the onboarding.

3.3 BEHAVIORAL ECOSYSTEM ENGAGEMENT

We evaluated point redemption patterns across customer segments to understand this churn risk.

	Loyalty Card Tier	Marketing Segment	Avg_Points_Redeemed	Avg_Churn_Risk	Total_Customer_Count
1	Aurora (Tier 3)	Lowest Value customers	83.9	96.81%	81
2	Aurora (Tier 3)	Only uses when discounts	643.2	6.58%	233
0	Aurora (Tier 3)	Best Customers	958.8	2.08%	358
4	Nova (Tier 2)	Lowest Value customers	116.0	94.59%	182
5	Nova (Tier 2)	Only uses when discounts	656.5	5.62%	384
3	Nova (Tier 2)	Best Customers	980.9	2.11%	582
7	Star (Tier 1)	Lowest Value customers	76.5	96.49%	201
8	Star (Tier 1)	Only uses when discounts	645.5	5.36%	582
6	Star (Tier 1)	Best Customers	962.2	2.18%	745

The data established a direct link between account activity and customer retention. Stable, low-risk cohorts show higher interaction with the rewards system, averaging over 950 points redeemed.

Conversely, high-risk, low-value cohorts show clear dormancy patterns with point redemptions dropping below 116 points. Once a customer stops redeeming points, they lose touch with our brand value proposition and enter a high-risk churn state.

4. OPERATIONAL RETENTION PLAYBOOKS

To ensure we can immediately work upon these insights, my automated segmentation logic categorizes at-risk accounts into 3 operational campaigns. The specifics of each of these campaigns are as follows:

4.1 CAMPAIGN ALPHA: EXPERIENTIAL UPGRADE

- **WHO:** Aurora cardholders who joined through partner promotions and show a model churn probability of $\geq 50\%$.
- **WHEN:** When the account's quarterly metric drops below 0.75.
- **WHY:** Premium travelers are rarely motivated by minor point discounts. They value status, efficiency, and comfort.
- **WHAT:** Direct priority outreach offering an unearned, complimentary premium airport lounge pass or an immediate business-class seat upgrade on their next route query.

4.2 CAMPAIGN BETA: HANDLING POINT HOARDERS

- **WHO:** Star and Nova promotional members showing high churn risk and dormant point balance.
- **WHEN:** After 90 consecutive days of complete flight inactivity.
- **WHY:** Hoarding points doesn't create any loyalty relationship. Getting a customer to redeem points just once builds ecosystem investment.
- **WHAT:** An automated mobile app push notification or personalized email alert stating something along the lines of: "Your points are expiring soon. Redeem now for a 0-transaction fee for a short weekend getaway."

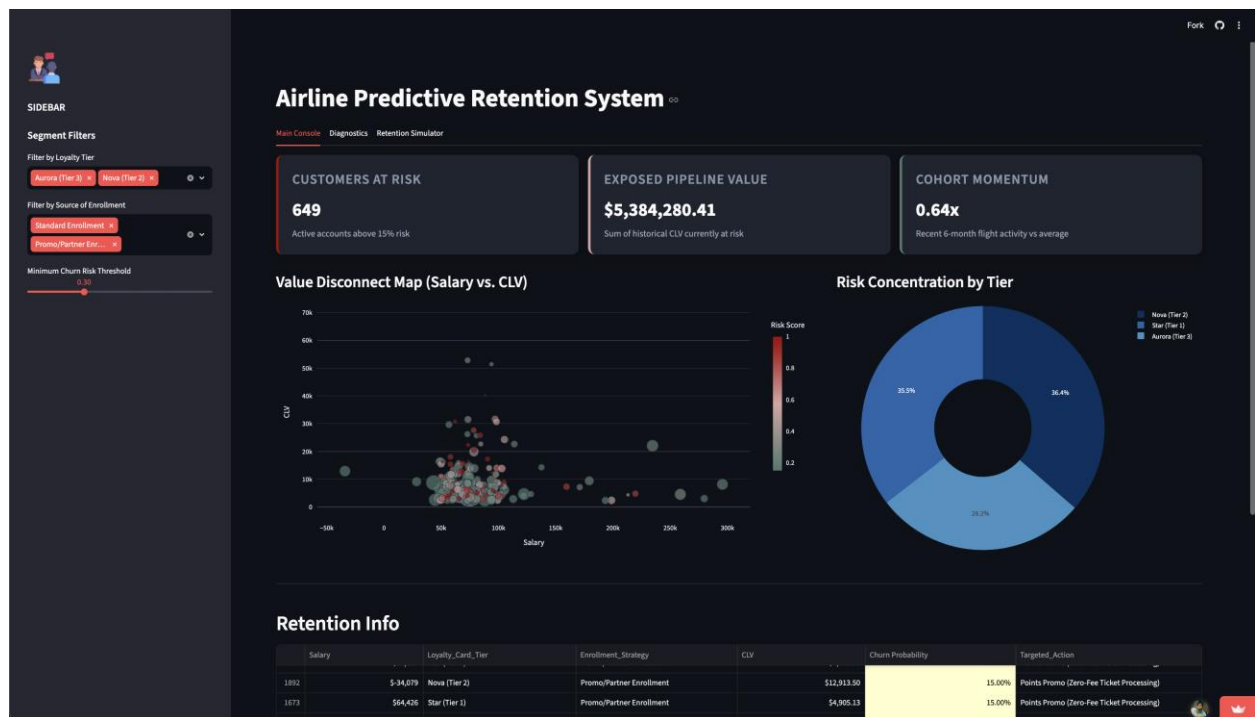
4.2 CAMPAIGN GAMMA: VIP GUESTS

- **WHO:** Organically enrolled members whose churn probability is currently below 20%.
- **WHEN:** Automatically deployed on the anniversary of their account creation milestone or if provided; on their birthdays or anniversary.
- **WHY:** Protecting our low-risk high-value core is much more cost effective than winning over lost accounts. It reinforces existing brand loyalty.
- **WHAT:** Low-cost physical gifts, early priority access to seasonal route releases, or unexpected bonus miles added to their accounts.

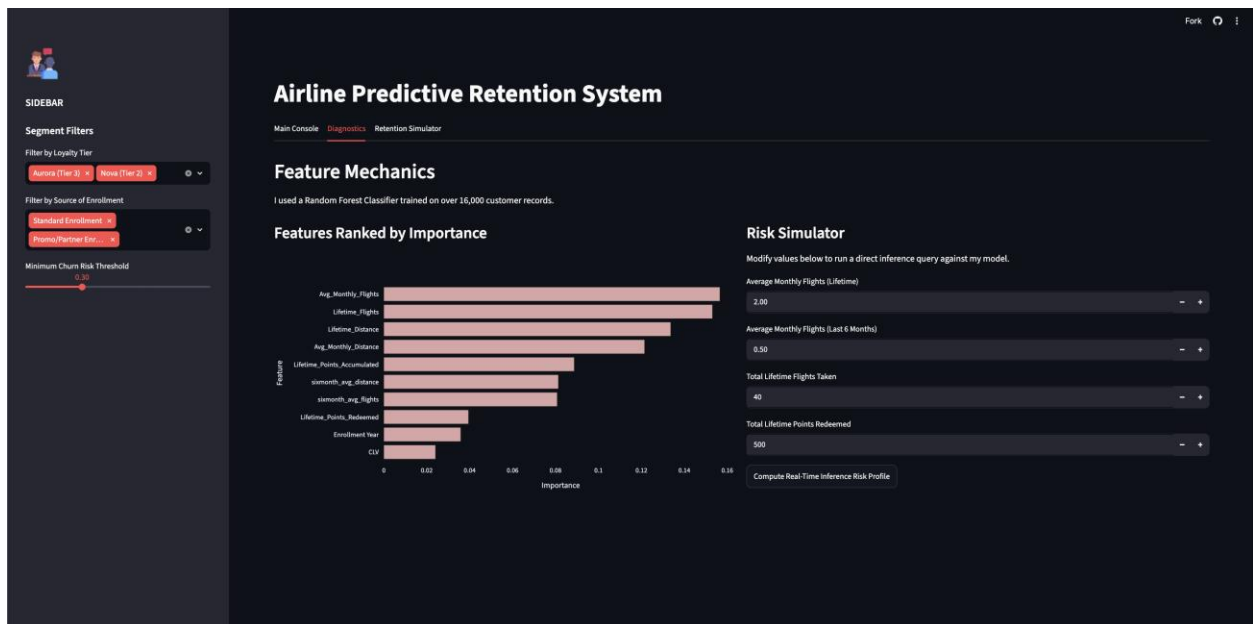
5. PROTOTYPE

To transform these models into easy-to-understand daily business operations, I built and deployed an interactive dashboard using Streamlit. The interface presents complex backend calculations via a simple and clean command center.

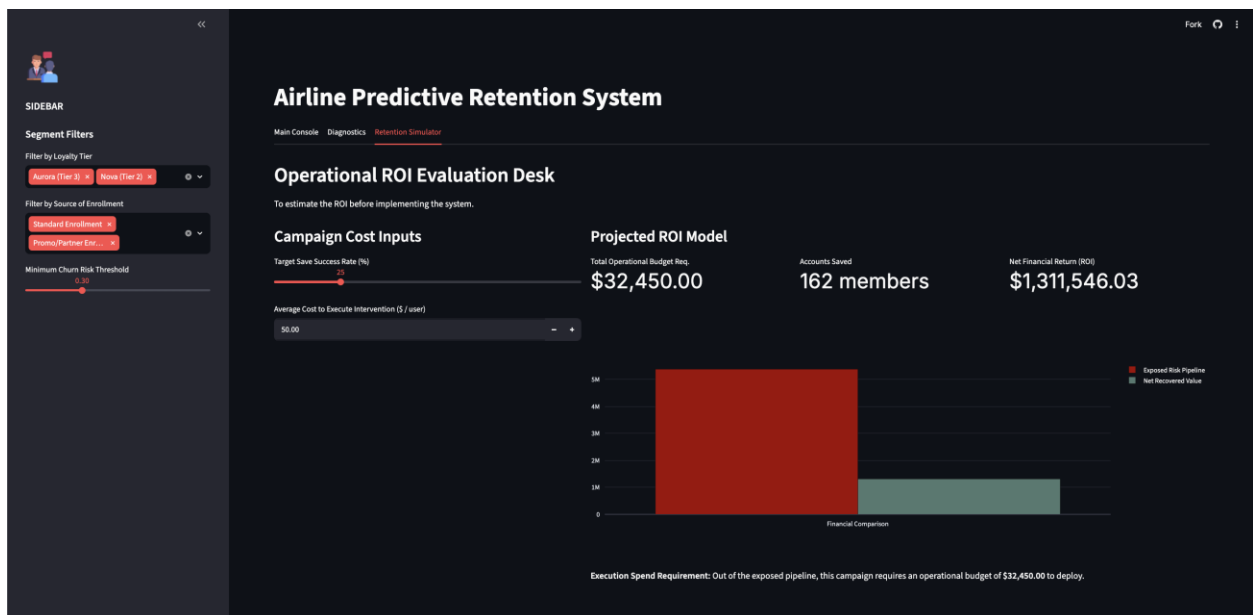
The deployed control center is structured into 3 structured dedicated workspaces to serve various functions:



MAIN CONSOLE: Provides managers with a view of the current financial exposure. Features an automated metric card displaying total at-risk customer counts and immediate revenue pipeline exposure.



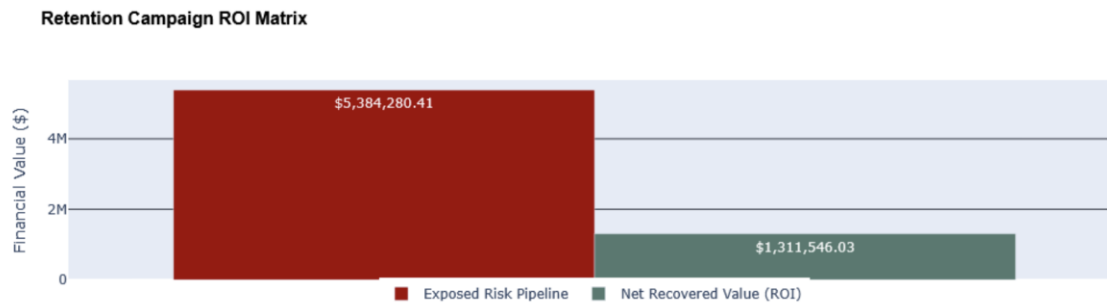
DIAGNOSTICS: Designed for technical analysts to review real-time model behaviors. Features an interactive “What-If” account risk simulator where managers can manually enter flight frequencies & other metrics to test individual customer risk scores.



RETENTION SIMULATOR: Designed for the marketing teams to model their finances before launching a campaign. Users can adjust conversion success rates and per-user execution costs to evaluate their budget requirements against the net pipeline saved.

6. ROI MODELLING

To verify the financial viability of the campaigns before committing the marketing spend, the system calculates via a ROI model by cross referencing the pipeline values against campaign implementation costs.



Execution Spend Requirement: Out of the exposed pipeline (649 accounts at risk), this campaign requires an operational budget of \$50.00 per user (\$32,450.00 total allocation) to deploy across the target segment.

EXECUTION BUDGET REQUIREMENT: This sample campaign requires an operational budget of \$50.00 per user to deploy across the filtered target segment.

By assessing financial metrics inside this deployment workspace, marketing teams can adjust intervention spends based on a cohort's average CLV. This ensures that every dollar spent on retention is mathematically optimized to protect our bottom line and maximize customer recovery.

7. CONCLUSION AND NEXT STEPS

The model shifts my retention strategy from tracking past losses to protecting future revenue. The data proves that relying on basic demographics and static tier-wise data leaves the business exposed to unexpected churn. Real financial protection requires monitoring active customer behavior and interaction trends.

Immediate Action Plan:

1. Deploy the Platform: Provide marketing and route operations teams access to the dashboard to embed the risk tracking feature provided into weekly planning cycles
2. Launch Campaign Alpha: Extract list of high-risk Aurora members from the platform and route them directly to the team for immediate outreach
3. Automate Campaign Beta instructions: Link the customer notification systems directly to the model's output data to trigger point-burning alerts the moment a promotional user's flight momentum slows down.

Airline can systematically reduce customer churn using this framework and optimize their marketing spending.